SIH 2026

PRODUCT REQUIREMENTS DOCUMENT

Food Traceability + Authenticity + Consumer Accountability + Risk Response Platform

Version 1.0 | *Requirements Baseline* | *14 August 2026*

Document purpose: Freeze what the team is building, why it is being built, what every major feature must do, how the components interact, and what belongs in the MVP/Base Model versus later phases.

1. Executive Summary

The proposed platform is a food traceability and safety system that creates a trusted digital journey for food products from source to consumer. Each relevant batch and, where required, individual consumer unit receives a digital identity. Supply-chain stakeholders append verified events rather than overwriting history. When products are transformed, parent-child lineage is preserved so the platform can trace an incident both backward to possible sources and forward to potentially affected products and locations.

The consumer-facing layer uses an outer QR for traceability and a proposed concealed/tamper-evident inner credential for an additional physical-authenticity check. The inner credential is deliberately not treated as a final cryptographic design yet; the source material states that its exact security construction must be designed and tested separately.

The user-provided concept extends this traceability foundation with a Consumer-Driven Feedback & Accountability Layer. Consumer complaints are associated with the relevant unit/batch and supply-chain context. A single complaint creates local accountability; repeated similar complaints across locations can trigger pattern detection and upstream escalation; verified systemic issues can support corrective action, alerts and targeted recall.

Product thesis

Identify the food → validate it → track its journey → preserve its lineage → verify the physical product → collect consumer feedback → detect risk → trace the impact → hold the right layer accountable → act.

2. Problem Statement

- Food supply-chain information is fragmented across farmers/producers, processors, manufacturers, transporters, retailers and other organizations.
- During a safety incident, organizations may struggle to reconstruct where a product came from, what was made from the same source, where affected products went, and which units or locations should be recalled.
- A visible QR can expose a valid digital history without proving that the physical package carrying the QR is the same item represented by that history.
- Consumer feedback is often disconnected from product lineage and therefore cannot easily become an operational safety signal.
- A single complaint may be isolated, while geographically distributed complaints against the same batch may indicate a systemic issue. The platform must distinguish between these cases.
- Businesses need more than provenance: they need a centralized view of product movement, quality events, consumer interaction, complaints, batch risk and supply-chain performance.

3. Product Goals & Non-Goals

Goals	Non-goals / explicit boundaries
End-to-end product/batch traceability	Do not make blockchain the whole application or a replacement for PostgreSQL.
Trusted multi-organization event history	Do not record every raw QR scan as a blockchain business event.
Batch and unit lineage	Do not require every kilogram of raw material to have an individual QR.
Consumer traceability + additional authenticity check	Do not claim the inner credential design is finalized before security analysis.
Bidirectional risk propagation	Do not force graph analytics or complex queries into chaincode.
Consumer feedback → accountability → escalation	Do not automatically declare a complaint true merely because it is recorded immutably.
Targeted recall and corrective-action support	Do not assume the system itself replaces regulatory recall procedures.
Business intelligence and supply-chain visibility	Do not expose commercial data to every participant by default.

4. Stakeholders, Roles & Permissions

Actor	Primary capabilities	Key restrictions
Farmer / Raw-material supplier	Provide/register source material information; attach evidence where applicable.	Only authorized operations; cannot alter historical events.
Producer / Processor	Validate incoming material; create/transform batches; record quality/process events.	Must respect product state and authorization rules.
Manufacturer	Receive, verify, transform, create final batches/units, block or initiate actions according to permissions.	Cannot receive blocked/invalid batches.
Transporter / Logistics	Receive/transfer products; record transport-related events and status.	Cannot perform manufacturing or ownership actions outside role.
Retailer	Receive inventory; associate product with sale/consumer interaction; respond to complaints.	Accountability is tied to relevant transactions/context; no upstream edits.
Consumer	Scan outer QR; perform inner verification where supported; view permitted traceability data; submit feedback/incident reports.	No access to restricted commercial records.
Manufacturer / Brand owner	Monitor end-to-end dashboard; analyze batches, complaints, scans, distribution and partners.	Access limited to authorized products/organizations/data.
Regulator / Authority	Review permitted traceability, incidents, evidence and recall scope; support enforcement/oversight.	Controlled access to sensitive evidence.
Platform administrator	Manage organizations, roles, configuration, thresholds, system health.	Administrative actions audited.

5. Functional Requirements

FR-01: Organization & Identity Management

- The system shall maintain organizations participating in the supply chain.
- Each user shall be associated with an organization and role.
- Authorization shall be checked before protected supply-chain operations.
- Important organization/user actions shall be auditable.
- The permission model shall support controlled visibility between organizations.

FR-02: Product, Batch/Lot & Unit Identity

- The system shall distinguish product definition, batch/lot, individual unit, organization, event, lineage relationship, evidence and risk incident.
- A valid incoming batch shall receive a unique digital identity.
- Unit-level identities shall be created where required, especially at manufacturing/packaging.
- QR codes shall reference system identities; the QR itself shall not be treated as the identity.
- The system shall support a structure such as Wheat W001 → Flour F001 → Biscuit B001 → B001-U0723.

FR-03: Supply-Chain Event Recording

- Stakeholders shall add meaningful business events such as registration, validation, receipt, transfer, transformation, blocking and recall.
- Events shall append to history; previous events shall not be overwritten.
- A QR scan shall be treated as an interaction, not automatically as a business event.
- Before a trusted event is committed, the system shall validate identity, organization/role, current product state and requested operation.
- Critical trusted events shall be recorded through Hyperledger Fabric chaincode and ledger mechanisms.

FR-04: Product Transformation & Lineage

- The system shall create parent-child relationships when material is processed or transformed.
- Lineage shall support one-to-many and potentially many-to-one transformations as required by the real process.
- The platform shall support backward traversal to source/parent material.
- The platform shall support forward traversal to derived batches, units, retailers and locations.
- Lineage shall be queryable outside the blockchain for efficient risk analysis.

FR-05: Outer QR Traceability

- The consumer shall be able to scan the visible outer QR.
- The traceability page shall present understandable product information, permitted provenance stages, relevant dates and current status/trust state.
- The QR shall resolve to the registered product/unit identity.
- A copied outer QR shall not be represented as sufficient proof of physical authenticity.

FR-06: Inner Credential / Authenticity Layer

- The product may contain a concealed or tamper-evident inner credential.
- The consumer shall be able to perform a second verification where supported.
- The system shall compare the presented credential with the registered cryptographic binding for the unit.
- The UI shall distinguish traceability success from physical-authenticity success.
- Missing, inconsistent or suspicious credentials shall generate a clear warning.
- The exact cryptographic construction, binding method and packaging/tamper-evidence design shall remain a dedicated security-design task.

FR-07: Consumer Feedback & Incident Reporting

- The consumer shall be able to report issues associated with the scanned product/unit.
- Report categories shall include, at minimum: damaged packaging, poor quality, suspected contamination, incorrect labeling, expired product, authenticity concern and other concern.
- The report shall capture the product/unit or batch identity, timestamp, report category, description and optional evidence such as images.
- The system shall associate the report with the relevant supply-chain context and transaction where available.
- The system shall provide a status lifecycle such as Submitted → Under Review → Escalated/Assigned → Resolved/Closed, with additional statuses allowed during implementation.
- Consumer reports shall be treated as risk signals, not automatically as verified facts.

FR-08: Consumer-Driven Accountability

- The nearest responsible supply-chain layer shall receive the first accountability signal for a consumer complaint when the transaction context supports that attribution.
- A retailer may therefore be the first accountable layer for a purchase made from that retailer.
- Accountability scoring/rating shall be based on complaint signals associated with relevant products/transactions and shall be auditable.
- The system shall avoid attributing upstream responsibility solely from a single consumer complaint when evidence does not support it.
- When a repeated pattern indicates an upstream issue, accountability shall expand/escalate to the relevant distributor, manufacturer, processor or supplier.
- The accountability model shall separate complaint volume, severity, verification status and confirmed/systemic incidents.

FR-09: Pattern Detection & Escalation

- The system shall aggregate similar complaints by product, batch, issue category, time window and geography.
- The engine shall distinguish isolated complaints from repeated or geographically distributed patterns.
- Configurable thresholds shall determine when an issue moves from local monitoring to upstream escalation.
- Escalation shall consider count, distribution, severity and verification status rather than count alone.

- Once a threshold is crossed, the system shall identify the next relevant upstream layer using product lineage and supply-chain relationships.
- Escalation shall generate an incident/risk record and notify authorized stakeholders.
- Thresholds shall be configurable so the pilot does not hard-code arbitrary values.

FR-10: Risk Propagator

- A consumer incident shall trigger backward traversal toward potential source material.
- The same incident shall support forward traversal to identify other potentially affected derived batches, units, retailers and locations.
- An upstream confirmed defect shall trigger downstream impact analysis.
- The engine shall produce an affected-scope set for investigation, alerting or targeted recall.
- Complex graph traversal shall be performed in the operational/query layer rather than inside Fabric chaincode.

FR-11: Recall, Block & Corrective Action Support

- Authorized stakeholders shall be able to mark relevant product/batch states such as blocked, recalled or under investigation.
- The platform shall generate a traceable affected scope from lineage and distribution data.
- The dashboard shall identify relevant products, batches, units, retailers and locations.
- Recall actions and status changes shall become auditable events.
- The system shall support targeted action rather than requiring a blanket market-wide recall.

FR-12: Evidence Management

- Certificates, laboratory reports, inspection reports, images and other large evidence shall be stored outside the blockchain in the evidence layer.
- IPFS shall be used as the proposed evidence store, with content identifiers/references or cryptographic commitments recorded in the trusted layer.
- Evidence shall be linked to products, batches, incidents or events.
- Access to sensitive evidence shall be permission-controlled.

FR-13: Business Intelligence & Company Dashboard

- Authorized businesses shall have a centralized view of the supply chain from source to consumer.
- Dashboard views shall include product/batch information, movement history, responsible stakeholder, quality/inspection records, consumer feedback, batch complaints, traceability events and distribution performance.
- The system shall provide complaint and risk analytics by batch and geography.
- The system shall provide scan analytics such as scan frequency, regional engagement, batch reach and geographic concentration of complaints.
- The dashboard shall help answer: Where is the product? How is it performing? Where are consumers engaging? Where are complaints occurring? Which batch is affected? Which supply-chain partner is associated? How is distribution performing?

FR-14: Consent-Based Location Intelligence

- Where location is collected from consumer scans, collection shall be consent-based and privacy-aware.
- The platform shall use location at an appropriate granularity for business intelligence rather than exposing unnecessary precise consumer location.
- Location analytics shall support product movement, regional engagement, complaint concentration and distribution optimization.
- Consumer-facing privacy disclosure and consent controls shall be part of the UX.

FR-15: Permissioned Data Visibility

- Consumers shall see only consumer-appropriate traceability and verification information.
- Supply-chain organizations shall see information permitted by their role and network membership.
- Commercially sensitive records shall not become globally visible merely because they exist in the system.
- The architecture shall allow stronger private-channel/selective-disclosure mechanisms in later phases.

6. End-to-End Core Workflows

A. Normal Supply-Chain Flow

1. Farmer supplies raw material.
2. Producer/processor validates and registers source batch.
3. System assigns batch identity and records the trusted registration event.
4. Processor/manufacturer receives and verifies the batch.
5. Transformation creates a child batch and preserves parent-child lineage.
6. Manufacturer creates final batch and, where required, individual unit identities.
7. Outer QR and inner credential are associated with the relevant unit.
8. Transporter records verified transfer events.
9. Retailer receives the product and records the receiving event.
10. Consumer scans the outer QR and optionally performs the inner authenticity check.

B. Consumer Complaint → Accountability → Escalation

11. Consumer scans the inner/consumer credential and the system identifies the unit/batch.
12. Consumer submits an issue report with category, description and optional evidence.
13. System validates the report context and creates an incident record.
14. Nearest responsible layer is assigned the initial accountability signal.
15. Accountability metrics are updated using severity, verification and relevant complaint history.
16. The system aggregates similar complaints against the same batch/product and across locations.
17. If the configured pattern threshold is reached, the incident is escalated upstream using lineage and supply-chain relationships.
18. Risk Propagator traverses upstream to potential sources and downstream to potentially affected products and locations.
19. Authorized stakeholders receive alerts and review the evidence.
20. The organization may block affected stock, initiate corrective action or create a targeted recall scope.
21. Incident status and final resolution are recorded for auditability.

C. Upstream Defect → Targeted Recall

22. Lab/manufacturer/regulator identifies a faulty source or batch.
23. Authorized user marks the relevant batch as under investigation/blocked/confirmed according to workflow.
24. Risk Propagator traverses downstream through lineage.
25. System identifies derived batches, units, destinations, retailers and relevant locations.
26. Dashboard generates an affected scope for action.
27. Recall/block/alert events are recorded.
28. Stakeholders can monitor resolution and close the incident.

7. Proposed System Architecture

Layer	Component	Responsibility
Experience	Consumer Web/Mobile UI	QR verification, traceability view, inner credential check, complaint reporting, consent.
Experience	Stakeholder Dashboard	Supply-chain operations, batch/lineage views, incidents, accountability, recall and analytics.
Application	Backend API	Authentication, orchestration, validation, permissions, workflow and service coordination.
Trusted Data	Hyperledger Fabric	Permissioned shared ledger for critical identities, states, events and commitments.
Trusted Logic	Fabric Chaincode	Authorization, valid state transitions, transfers, transformations, blocked-state and critical business rules.

Operational Data	PostgreSQL	Products, batches, units, organizations, lineage, incidents, scan logs, analytics results and recall sets.
Performance	Redis	Cache for frequent QR lookups, public history and risk summaries; never source of truth.
Evidence	IPFS	Certificates, lab reports, inspection reports, images and other large evidence.
Safety Intelligence	Risk Propagator	Bidirectional graph traversal and affected-scope computation.
Identity Layer	Outer QR + Inner Credential	Public traceability reference plus proposed additional physical-authenticity verification.

Architecture principle: each technology exists for a specific responsibility; the platform is not a blockchain-only application.

8. Core Data Model

Entity	Purpose	Key relationships
Organization	Known participant in network.	Has users/roles; owns or handles products.
User / Role	Identity and permissions.	Belongs to organization; authorized for operations.
Product Definition	Defines what the product is.	Referenced by batches/units.
Batch / Lot	Traceability grouping for a quantity.	Parent/child lineage; events; units.
Unit	Individual consumer package where needed.	Belongs to batch; QR/credential; incidents.
Event	Meaningful business occurrence.	References actor, product/batch/unit and timestamp.
Lineage Relationship	Parent-child transformation relationship.	Connects source and derived batches/units.
QR Credential	Reference/verification credential.	Bound to unit or product identity.
Incident	Complaint, contamination, quality or safety signal.	Linked to unit/batch, location, evidence and workflow.
Evidence	Supporting document/media.	Linked to event/incident/product/batch.
Accountability Record	Responsibility/performance signal.	Linked to stakeholder + incident/product context.
Risk/Recall Set	Computed impact scope.	Contains affected batches/units/locations/stakeholders.

9. Accountability Model

The accountability layer must be designed as a signal-and-escalation system, not a simplistic star-rating system.

Signal level	Meaning	System response
Level 1 — Local complaint	One or limited complaint tied to a transaction/unit.	Assign initial responsibility to nearest relevant layer; open review.
Level 2 — Repeated pattern	Similar complaints for same product/batch or relevant issue class.	Increase risk signal; notify responsible organization.
Level 3 — Distributed batch pattern	Similar complaints across multiple locations/transactions.	Trigger configurable escalation review; propagate upstream.
Level 4 — Verified systemic issue	Evidence/review confirms a batch or process problem.	Block/recall/corrective action workflow; downstream impact analysis.

Important design rule: complaint count alone must not prove fault. Severity, similarity, geographic distribution, verification status, evidence and lineage context should influence escalation.

10. Dashboard Requirements

Dashboard	Minimum widgets / views
Supply Chain Overview	Source → Processing → Manufacturing → Distribution → Retail → Consumer; current state; exceptions.
Batch Detail	Batch identity, parent/child lineage, events, quality records, current status, units, destinations.
Consumer Intelligence	Scan volume, engagement by region, batch reach, complaint

	concentration, trend over time.
Accountability	Complaints by stakeholder/layer, open incidents, repeated issues, escalation state, resolution time.
Risk & Incidents	Severity, affected scope, upstream/downstream graph, evidence, alerts, investigation status.
Recall Center	Affected products/units/locations, recall scope, block status, actions and completion.
Evidence	Certificates, lab/inspection reports, linked evidence and access state.
Audit Trail	Who performed what action, when, against which product/batch/unit.

11. Non-Functional Requirements

- Integrity: Critical events and state transitions must be tamper-evident and auditable.
- Authorization: Protected operations must be role- and organization-aware.
- Privacy: Consumer and commercial data must be exposed only at the necessary level; location collection must be consent-based.
- Availability: Consumer QR verification should remain usable even when advanced analytics services are unavailable.
- Performance: Frequently accessed public history and QR lookups should use caching where appropriate.
- Scalability: Lineage and incident analysis should be designed to support large batch/unit graphs without moving complex computation into chaincode.
- Observability: API calls, scans, errors, critical state changes, incidents and escalations should be logged and monitored.
- Auditability: Important business events, accountability changes and recall actions must be traceable.
- Modularity: EPCIS-compatible events, IoT, streaming, stronger privacy and AI/ML should be addable without rebuilding the core.

12. Security, Privacy & Trust Requirements

- Use a permissioned identity model for known organizations.
- Enforce critical authorization/state rules through chaincode where shared trust is required.
- Keep large evidence off-chain and preserve integrity references/commitments in the trusted layer.
- Do not equate immutability with truth: input validation, evidence and trusted data sources remain necessary.
- Do not claim that two different hash algorithms automatically prevent counterfeiting.
- The inner credential must be cryptographically bound to the registered unit and paired with a suitable tamper-evident physical design before it is treated as an authenticity mechanism.
- Consumer location data must be collected only with appropriate consent and used at a proportionate granularity.
- Sensitive organizational data should be permissioned and, where required later, separated using private channels/selective disclosure.

13. MVP / Base Model Scope

Priority	Feature	MVP status
P0	Organizations, roles and authentication	Must have
P0	Product, batch and unit identity	Must have
P0	Parent-child lineage	Must have
P0	Lifecycle states	Must have
P0	Fabric + chaincode for critical rules/events	Must have
P0	PostgreSQL operational/lineage database	Must have
P0	Outer QR traceability	Must have
P0	Supply-chain event recording	Must have
P0	First Risk Propagator version	Must have
P0	Consumer incident/feedback reporting	Must have
P1	Accountability scoring + configurable escalation thresholds	High priority
P1	Batch-level complaint pattern detection	High priority

P1	Basic business dashboard	High priority
P1	Redis caching	Base architecture / performance
P1	IPFS evidence storage	Base architecture / evidence
P1	Basic recall/block workflow	High priority
P2	Inner authenticity mechanism	Focused security module after base identity/lineage
P2	EPCIS-compatible event structures	Later phase
P2	IoT sensors / event streaming	Later phase
P2	Advanced privacy/selective disclosure	Later phase
P2	AI/ML risk analytics	Later phase
P2	Advanced hardware authenticity	Later phase

14. Acceptance Criteria / Definition of Done

- A sample food product can be traced from raw-material batch to final consumer unit.
- Every meaningful stage appends a valid event without overwriting prior history.
- A transformed product retains parent-child lineage.
- A consumer can scan the outer QR and see an understandable traceability record.
- A consumer can submit a complaint tied to the correct unit/batch.
- The system assigns the initial complaint to the nearest relevant accountability layer.
- Repeated complaints for the same batch can be aggregated and analyzed by issue category and geography.
- A configurable threshold can trigger escalation to an upstream layer.
- Risk Propagator can move upstream to potential sources and downstream to potentially affected products/locations.
- An authorized stakeholder can identify a targeted recall/block scope from the lineage graph.
- Critical events and state transitions are recorded through the trusted layer.
- Operational data remains queryable in PostgreSQL; Redis is used only as cache.
- Large evidence is stored through the evidence layer and linked to relevant incidents/events.
- Consumer-facing traceability and physical-authenticity status are presented as distinct claims.
- Location intelligence is consent-based and does not expose unnecessary consumer precision.
- The system can distinguish an unverified complaint from a verified/systemic issue.

15. Requirements Verification Matrix

User requirement / concept	Covered by	Verification
Complete food journey from farmer to consumer	FR-02, FR-03, FR-04, workflow A	PASS — explicitly covered.
Two QR/credential concept	FR-05, FR-06	PASS — outer traceability + inner authenticity preserved.
Consumer can report damaged/poor/contaminated/label/expiry issues	FR-07	PASS — categories explicitly included.
Nearest layer initially accountable	FR-08 + workflow B	PASS — with attribution caveat.
Multiple complaints trigger pattern detection	FR-09	PASS — batch/time/geography aggregation.
Escalation upstream to distributor/manufacturer/processor/supplier	FR-08, FR-09, FR-10	PASS.
Batch-wide issue → corrective action/recall	FR-11 + risk workflow	PASS.
Consumer location for business intelligence	FR-14 + FR-13	PASS — consent/privacy constraint added.
Company sees complete chain in one dashboard	FR-13	PASS.
Business questions: where/what/how/which batch/which partner	FR-13	PASS.
Blockchain + Fabric + chaincode	Architecture + FR-03	PASS.
PostgreSQL + Redis + IPFS	Architecture + FR-12	PASS.
Bidirectional Risk Propagator	FR-10	PASS.
Targeted recall	FR-11	PASS.
Do not put everything on blockchain	Architecture + boundaries	PASS.

Authenticity caveat: hash algorithms alone are insufficient	FR-06 + security	PASS.
Base Model before advanced IoT/AI/EPCIS	MVP scope	PASS.

Verification conclusion: The PRD covers all major requirements explicitly present in the supplied concept and source documents, while preserving the source documents' boundaries around blockchain scope, authenticity design, privacy and future-phase technologies. The main additions are formal product requirements for consumer accountability, configurable escalation, complaint verification, consent-based location intelligence and acceptance criteria so the concept can be implemented and tested.

16. Reference Scenario — Biscuit Packet

Wheat Batch W001 is registered → Flour Batch F001 is derived → Biscuit Batch B001 is produced → 10,000 consumer units are created. A unit such as B001-U0723 reaches a retailer. The consumer scans the outer QR, verifies traceability, optionally performs the inner check and reports suspected contamination. The incident is associated with U0723 and B001. The nearest relevant retail layer receives the initial accountability signal. If similar reports arise from multiple locations for B001, the pattern detector escalates the issue. The Risk Propagator follows B001 → F001 → W001 upstream and also checks sibling/descendant batches and units downstream. Authorized stakeholders can then investigate, block affected stock or create a targeted recall scope.

17. Success Metrics

Metric	Why it matters
Traceability completeness	Percentage of demo/production products with complete required event chains.
Lineage reconstruction success	Ability to reconstruct upstream/downstream relationships for an incident.
Incident-to-impact time	Time required to identify affected batches/units/locations.
Complaint classification accuracy	Quality of mapping reports into issue categories.
Escalation precision	How often escalations correspond to genuinely relevant batch/process signals.
Recall scope precision	Ability to identify targeted affected products rather than broad unnecessary scope.
Consumer verification success	Successful traceability and authenticity-check completion.
Dashboard usefulness	Time for authorized users to answer core business/safety questions.
Audit completeness	Percentage of critical state changes with attributable records.

18. Recommended Implementation Order

29. Freeze entities, roles, lifecycle states and business rules.
30. Implement organization/user authentication and authorization.
31. Implement product, batch, unit and lineage data model.
32. Implement PostgreSQL operational schema.
33. Implement Fabric network/gateway and focused chaincode.
34. Implement verified supply-chain event flows.
35. Implement outer QR generation/lookup and consumer traceability page.
36. Implement consumer feedback/incident reporting.
37. Implement evidence upload/reference handling.
38. Implement first Risk Propagator for upstream/downstream traversal.
39. Implement accountability signals and configurable escalation thresholds.
40. Implement basic incident, recall and business dashboards.
41. Add Redis caching and performance optimization.
42. Design and test the inner authenticity mechanism as a separate security module.
43. Only then extend to EPCIS, IoT, streaming, advanced privacy and AI/ML.

19. Key Risks & Mitigations

Risk	Mitigation
Garbage-in, garbage-out	Validation, authorization, evidence, quality checks and future trusted automated data sources.
Copied outer QR	Treat outer QR as traceability reference; use additional concealed/tamper-evident credential.
False consumer complaints	Verification workflow, evidence, severity and pattern-based escalation; do not equate complaint with proof.
Over-attribution to retailer	Initial accountability is contextual; systemic escalation follows evidence and lineage.
Blockchain becomes too heavy	Store operational data in PostgreSQL, evidence in IPFS and cache in Redis; keep chaincode focused.
Privacy leakage	Permissioned access, consent-based location and appropriate data granularity.
Recall too broad/narrow	Bidirectional lineage + computed affected scope + human review before final action.
Complex graph at scale	Queryable lineage model in PostgreSQL and dedicated Risk Propagator.
Premature cryptographic design	Separate inner-credential security design and test attack scenarios before freezing construction.

20. Glossary

Term	Meaning
Blockchain	Shared, tamper-evident ledger maintained by authorized participants.
Hyperledger Fabric	Permissioned blockchain framework suited to known organizations and enterprise workflows.
Chaincode	Fabric smart-contract logic used to enforce trusted business rules.
Lineage	Parent-child relationships connecting raw materials, derived products, batches and units.
Batch/Lot	Quantity of product treated as a traceability group.
Unit	Individually identifiable consumer package/item.
Event	Meaningful business occurrence such as registration, receipt, transfer, transformation or recall.
Risk Propagator	Engine that traces a safety incident backward to possible sources and forward to potentially affected products/locations.
Authenticity binding	Cryptographic relationship connecting product/unit identity with the additional physical credential.
GIGO	Garbage In, Garbage Out; immutable storage does not make incorrect input correct.

21. Source Basis & Traceability to Supplied Documents

This PRD was prepared from the user's supplied Consumer-Driven Feedback & Accountability Layer and Business Perspective requirements, together with the supplied SIH 2026 Food Traceability Core Concept and Hinglish Core Story documents. The core documents establish the product identity, lifecycle events, lineage, bidirectional Risk Propagator, outer QR, proposed inner credential, Hyperledger Fabric/chaincode, PostgreSQL, Redis, IPFS, permissioned participation, consumer experience and phased Base Model approach.

The source documents explicitly state that the next engineering document should freeze entities, database schema, Fabric assets, chaincode functions, transaction flows, API contracts and implementation order; this PRD is intended to serve as that product-level requirements baseline before detailed engineering specifications.

Source reference: SIH 2026 Food Traceability System — Core Concept, Problem Understanding, Solution Story and Base Model. The document defines the platform as a food safety and traceability system rather than a blockchain application and emphasizes trusted journey, lineage and actionable risk response. □filecite□turn0file0□L13-L28□

Source reference: SIH 2026 Food Traceability System — Core Story • Solution • Base Model. The supplied story covers the farmer-to-consumer flow, lineage, Risk Propagator, inner credential, Fabric, PostgreSQL, Redis, IPFS and phased Base Model. □filecite□turn0file1□L12-L21□

22. Final Product Definition

We are building a trusted food journey platform that connects product identity, supply-chain events, product lineage, consumer verification, consumer feedback, accountability, risk propagation and business intelligence. The system should not merely answer “Where did this product come from?” It should also answer “What happened to it, where is the problem, who is initially responsible, what else may be affected, and what action should follow?”